EXPERIMENT 10: KNOWLEDGE MANAGEMENT SYSTEM

Aim:

To store and retrieve frequently asked questions and answers using a small searchable knowledge base.

Software Required: Python 3.x

Procedure / Algorithm:

1. Create a dictionary of topics and answers.
2. Accept a search topic from the user.
3. Display the matching answer or a not-found message.

Program:

```
knowledge_base = {
    "python": "Python is a high-level programming language.",
    "ontology": "An ontology formally defines concepts and relationships.",
    "knowledge graph": "A knowledge graph connects entities through relationships.",
    "expert system": "An expert system uses rules and knowledge to solve problems."
}

print("Knowledge Management System")
print("Available topics:", ", ".join(knowledge_base.keys()))

query = input("\nEnter a topic to search: ").strip().lower()

if query in knowledge_base:
    print("Knowledge Found:")
    print(knowledge_base[query])
else:
    print("No information found for this topic.")
```

Sample Input:

Enter a topic to search: ontology

Expected Output:

```
Python 3.12.2 (tags/v3.12.2:6abddd9, Feb  6 2024, 21:26:36) [MSC v.1937 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=== RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-10.py ===
Knowledge Management System
Available topics: python, ontology, knowledge graph, expert system

Enter a topic to search: python
Knowledge Found:
Python is a high-level programming language.
>>>
```

Result:

Thus, the program stored and retrieved knowledge using a searchable dictionary.